B3.3 Muscle and motility (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 DISTINGUISH between motile and sessile organisms, with an example of each. [B3.3.1] [2]

2 NAME the proteins in the thick and thin filaments of a sarcomere. [B3.3.2] [2]

3 DEFINE motor unit. [B3.3.4] [1]

4 STATE the roles of cartilage and synovial fluid in the hip joint. [B3.3.6] [2]

5 NAME the two bones that meet at the hip joint. [B3.3.6] [1]

6 STATE two reasons for locomotion, with an example of each. [B3.3.9] [2]

Section B: Understanding

14 MARKS

7 EXPLAIN the roles of calcium ions and ATP in the contraction of a sarcomere. [B3.3.2] [4]

8 EXPLAIN how the internal and external intercostal muscles act as an antagonistic pair. [B3.3.3, B3.3.8] [3]

9 **EXPLAIN** how a bone acts as a lever. [B3.3.5] [2]

10 **COMPARE** the range of motion at the hip and the knee. [B3.3.7] [2]

11 **EXPLAIN** three adaptations of marine mammals for swimming. [B3.3.10] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [B3.3.7]

Students measured the range of hip flexion (raising the straight leg forward while lying on their back) with a goniometer, before and after 10 minutes of stretching. Each value is the mean of three readings.

GROUP	NUMBER	BEFORE / °	AFTER / °
STUDENTS WHO PLAY SPORT WEEKLY	12	82 ± 6	91 ± 5
STUDENTS WHO DO NOT	12	68 ± 9	75 ± 8

a **CALCULATE** the percentage increase in range after stretching for the sport group. [2]

b **COMPARE** the two groups. [2]

c **SUGGEST** why each value is the mean of three readings. [1]

d **STATE** two variables that should be controlled when measuring. [2]

e **EVALUATE** the claim that playing sport causes a greater range of motion. [3]

Section D: Longer answer

6 MARKS

13 **EXPLAIN** how a skeletal muscle fibre contracts, using the sliding filament model. [B3.3.2, B3.3.4] [6]